

job access. Over repeated hiring cycles, such dynamics can create a “lock-in” effect, where the stylistic patterns of the dominant LLM become entrenched in applicant pools and further reinforce its advantage. For employers, this presents a double-edged challenge: while LLM-based screening promises efficiency and a more comprehensive assessment than traditional keyword matching, it simultaneously increases the risk of overlooking highly qualified candidates who do not use the “right” AI tool, while advancing less-qualified candidates whose resumes happen to align stylistically with the evaluator.

6. Bias Mitigation

Having established both the prevalence and labor market impact of self-preference bias in LLM-based resume evaluations, we next turn to potential remedies. Left unaddressed, this bias can distort hiring outcomes by systematically advantaging candidates who use the same LLM as employers, while disadvantaging equally qualified applicants who do not. Such dynamics raise fairness concerns for job seekers and pose risks for employers, who may inadvertently overlook strong candidates. In practice, employers are likely to seek mitigation tools that are simple, cost-effective, and compatible with existing screening workflows. Motivated by evidence that self-preference bias is linked to LLM models’ ability to recognize their own outputs, we evaluate two intervention strategies that directly target this self-recognition mechanism.

6.1. Mechanism: Self-Recognition

Recent work suggests that LLMs may possess an implicit ability to recognize text they have generated, and that this capacity is linked to self-preference. Benchmark datasets designed to probe situational awareness show that models can identify aspects of their own outputs and contexts (Laine et al. 2024). LLMs can also reliably distinguish their own generations from those of alternative models, with higher self-recognition capability often correlated with stronger self-preferencing biases (Panickssery et al. 2024). In addition, larger models appear to exhibit greater self-recognition, which may help explain why they show stronger self-preference in our empirical analysis (Laine et al. 2024, Panickssery et al. 2024). Taken together, these findings suggest that self-recognition is a plausible mechanism contributing to the bias we document. In the following, we evaluate mitigation strategies designed to directly target this mechanism.

6.2. System Prompting

Our first mitigation strategy addresses self-preference by disrupting the evaluator’s tendency to rely on stylistic or linguistic cues that signal its own outputs. Specifically, we modify the evaluator’s system prompt to explicitly discourage source-based judgments and instead focus attention on substantive content quality. For example, the revised prompt instructs: “You should not consider

Table 2 LLM-vs-Human Self-Preference Bias Before and After Mitigation

Bias Measure	GPT-4o	LLaMA-3.3-70B	DeepSeek-V3
<i>Before Mitigation</i>			
Self-Preference Bias (%)	82	79	72
<i>(1) After Mitigation via System Prompting</i>			
Self-Preference Bias (%)	61	30	60
Absolute Decrease (pp) ↓	21	49	12
Relative Decrease (%) ↓	26	62	17
<i>(2) After Mitigation via Majority Voting</i>			
Self-Preference Bias (%)	30	23	29
Absolute Decrease (pp) ↓	52	56	43
Relative Decrease (%) ↓	63	71	60

Notes: This table reports the LLM-vs-Human self-preference bias for three models before and after applying two mitigation strategies: (1) system prompting and (2) majority voting. Absolute decreases are reported in percentage points. Relative decreases are calculated as the percent reduction from the pre-mitigation bias.

or infer whether the resumes were written by a human or by AI. Focus only on the quality of the content.” This intervention aims to weaken the influence of self-recognition cues that drive models to favor their own generative style.

Applying this prompting strategy across evaluator models leads to consistent reductions in self-preference bias. For example, GPT-4o’s LLM-vs-Human bias decreases from 82% to 61%, and LLaMA-3.3-70B’s from 79% to 30%, after controlling for resume quality (Table 2). These results indicate that the bias is not hardwired into model architecture but can be shaped by context and instruction, providing evidence that explicit prompts can partially disrupt self-recognition.

6.3. Majority Voting Ensemble

Our second strategy mitigates self-preference bias through ensemble evaluation. Instead of relying on a single LLM to judge a resume pair, we construct a panel of three models: the target evaluator and two smaller models (LLaMA-3.2-1B and LLaMA-3.2-3B) that exhibit minimal self-preference. The final decision is determined by majority vote. This approach is motivated by recent evidence that stronger self-recognition ability is associated with greater self-preference bias (Panickssery et al. 2024). By combining models with weaker self-recognition tendencies, the ensemble leverages model diversity to dilute the bias of larger evaluators.

This mitigation strategy proves highly effective. Across all three LLM models tested, the average LLM-vs-Human comparisons self-preference bias can drop by over 50%. For example, as it is shown in Table 2, GPT-4o’s bias is reduced from 82% to 30%, and LLaMA-3.3-70B’s from 79% to 23%. These results demonstrate that both system prompting and majority voting offer robust and scalable approaches to mitigation, particularly in high-stakes applications where fairness is critical.

In summary, the two mitigation strategies demonstrate that self-preference bias, while widespread, is not immutable. With relatively simple design interventions, we can substantially reduce bias in LLM-based evaluations without modifying the underlying model weights or retraining. These findings offer a practical path forward for deploying LLMs in evaluative roles while minimizing unintended algorithmic unfairness.

7. Concluding Remarks

Our study documents a systematic form of algorithmic bias, AI self-preference, in the context of algorithmic hiring. Across two fairness metrics, we observe strong and consistent evidence of LLM-vs-Human self-preference in nearly all models tested. Simulation experiments show that in realistic hiring pipelines, candidates using the same LLM as the evaluator LLM are 23% to 60% more likely to be shortlisted than if they submit human-written resumes, with the largest disadvantage observed in business-related occupations such as sales and accounting. To address this issue, we propose two mitigation strategies (system prompting and majority voting) that firms can adopt with minimal implementation costs. Both approaches substantially reduce bias, cutting self-preference by more than half.

Beyond technical remedies, our findings have important policy implications. Current discussions of AI fairness largely focus on demographic disparities, but our results highlight the need to address biases arising from AI–AI interactions. Regulators and hiring platforms should recognize AI self-preference as a distinct and emerging form of algorithmic bias. Transparency requirements could mandate that organizations disclose whether AI is used in resume screening and what safeguards are in place to ensure fairness. In addition, third-party audits could incorporate self-preference metrics into fairness evaluations of AI-assisted hiring systems. Such measures would promote more accountable and equitable deployment of AI in employment contexts.

Finally, our study opens several directions for future research. As AI tools become more widely adopted, interactions between AIs will become increasingly common. Extra caution is warranted in contexts where AI systems are placed in evaluative or adjudicative roles, such as content moderation, grading in higher education, or other settings where they act as evaluator, judges or mediators. Another important direction is to examine how self-preference bias manifests in multilingual or cross-cultural environments, where non-English content may be especially vulnerable due to tokenization artifacts or limited representation in training data. Last but not the least, further investigation into the mechanisms underlying self-preference is needed. Rigorous study of self-recognition and other contributing processes will be critical to addressing the root causes of this bias and ensuring the fair integration of AI into hiring and other high-stakes decision domains.